

Need of Constraints and Its Types

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Review of previous session

In previous session we have learned to:

- ▶ Create basic table, i.e., STUDENT
- ▶ Perform insert operation on table
- ▶ Perform update operation on table
- ▶ Perform retrieve and delete operation on table.

This basic STUDENT table suffers from anomalies.



Issues in basic STUDENT table

Basic table suffers from following anomalies:

- ▶ There is no check on storing duplicate records
- ▶ There is no mandatory column
- ▶ There is no check on validity of data

To ensure all these features, there is a need to apply constraints during creation of table.



Constraints on a Table

- ▶ Constraints are rules enforced on the data in a database table to maintain the integrity, accuracy, and reliability of the data.
- ▶ They define conditions that must be met for the data to be valid and help ensure the consistency of relationships between tables in a relational database.
- ▶ Levels of Constraints?



Levels of Constraints

Column level constraints

Constraints which are applied on single column of table are known as column level constraints.

For example:

ID is Unique.

Table level constraints

Constraints which are applied on more than one column of a table are known as table level constraints.

For example:

ID and Degree combination is unique in primary School database

[In primary school Roll number of students starts with one in each Degree]



Types of Constraints

- ▶ Not Null
- ▶ Unique
- ▶ Primary key
- ▶ Check
- ▶ Foreign key
- ▶ Default



Not Null, Unique and Primary Key Constraint

Review of previous session

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- ▶ Perform insert operation on table
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This basic STUDENT table suffers from anomalies.



Null/Not Null Constraint

- ▶ Specifies if the column must contain value or might not contain any.
- ▶ By default all columns in a table allow nulls.
- ▶ NOT NULL specifies that all rows in the table to have value for specified column.
- ▶ Syntax:
Columnname datatype (size) [constraint constraintname]
NOT NULL



Example [Not Null: Column level constraint]

```
CREATE TABLE student
```

```
(
```

```
ID Number(4) Not Null,
```

```
Name Char(15), Degree Char(10), Marks Number(4),
```

```
DOB Date
```

```
);
```



Unique Constraint

- ▶ Enforce uniqueness of the column or group of columns.
- ▶ Syntax:
Columnname datatype (size) [constraint constraintname]
UNIQUE



Example [Unique: Column level constraint]

```
CREATE TABLE student
(
  ID Number(4) Unique,
  Name Char(15), Degree Char(10), Marks Number(4),
  DOB Date
);
```

OR

```
CREATE TABLE student
(
  ID Number(4) Constraint RNO_Unique Unique,
  Name Char(15), Degree Char(10), Marks Number(4),
  DOB Date
);
```



Example [Unique:Table level constraint]

```
CREATE TABLE primary_student  
(  
  ID Number(4), Name Char(15), Degree Char(10),  
  Marks Number(4), DOB Date,  
  Unique(ID , Degree )  
);
```

OR

```
CREATE TABLE primary_student  
(  
  ID Number(4), Name Char(15), Degree Char(10), Marks Number(4), DOB Date,  
  Constraint ID_Degree _Unique Unique(ID , Degree )  
);
```



Primary Key Constraint

- ▶ It is a combination of both the unique key constraint as well as the not null constraint.
- ▶ A primary key is used to identify each row of the table uniquely.
- ▶ A primary key may be either consists of a single field (Simple key) or group of fields (Composite Primary Key).
- ▶ Oracle enforces all PRIMARY KEY constraints using indexes.
- ▶ Syntax:
Columnname datatype (size) [constraint constraintname] PRIMARY KEY



Example [Primary Key: Column level Constraint]

```
CREATE TABLE student  
(  
  ID Number(4) Primary Key,  
  Name Char(15), Degree Char(10), Marks Number(4),  
  DOB Date  
);
```

OR

```
CREATE TABLE student  
(  
  ID Number(4) Constraint RNO_PK Primary Key,  
  Name Char(15), Degree Char(10), Marks Number(4),  
  DOB Date  
);
```



Example [Primary Key: Table level Constraint]

```
CREATE TABLE primary_student  
(  
ID Number(4), Name Char(15), Degree Char(10),  
Marks Number(4), DOB Date,  
Primary Key(ID , Degree )  
);
```

OR

```
CREATE TABLE primary_student  
(  
ID Number(4), Name Char(15), Degree Char(10), Marks Number(4), DOB Date,  
Constraint ID_Degree _PK Primary Key(ID , Degree )  
);
```



Summary

We have learned:

- ▶ Null/Not Null constraint
- ▶ Unique constraint
- ▶ Primary Key constraint

Next Session, we will cover:

- ▶ Check constraint
 - ▶ Foreign key
 - ▶ Default constraint
-



Default Constraint and Information of Constraints

Default Constraint

- ▶ The default value constraint allows the user to insert the values in the columns where the user do not want to insert the value.
- ▶ This is not actually a constraint and used to insert default value if value is missing for this column.
- ▶ The datatype of the default value should match the datatype of the column.
- ▶ Syntax:
- ▶ Columnname datatype (size) default value



Example [Default: Column level constraint]

```
CREATE TABLE student  
(  
  ID Number(4) Primary key,  
  Name Char(15),  
  Degree Char(10),  
  Marks Number(4) default 60,  
  DOB Date  
);
```



Objectives of this Session

We will discuss:

- ▶ Check constraint
- ▶ Demonstration of its implementation
- ▶ Limitations of check Constraint



Check Constraint

Check Constraint

- ▶ Check constraints allow Oracle to verify the validity of data being entered on a table against a set of constants. These constants act as valid values.
- ▶ The Check constraint consists of the keyword CHECK followed by parenthesized conditions.
- ▶ Check constraints must be specified as a logical expression that evaluates either to TRUE or FALSE.
- ▶ Syntax:
- ▶ Columnname datatype (size) [constraint constraintname] CHECK (logical expression)



Example [Check: Column level constraint]

```
CREATE TABLE student
```

```
(  
  ID Number(4) , Name Char(15),  
  Degree Char(10) Check (Degree in ('BS','MS','MBA')),  
  Marks Number(4) Check (marks>=0), DOB Date  
);
```

OR

```
CREATE TABLE student
```

```
(  
  ID Number(4) , Name Char(15),  
  Degree Char(10) Constraint Degree _check Check (Degree in ('BS','MS','MBA')),  
  Marks Number(4) Check (marks>=0), DOB Date  
);
```



Example [Check: Table level constraint]

```
CREATE TABLE library  
(  
  ID Number(4) Primary Key,  
  Book_Number Number(10) NOT NULL,  
  DOI Date, DOR Date,  
  Check (DOR>DOI);
```

OR

```
CREATE TABLE library  
(  
  ID Number(4) Primary Key,  
  Book_Number Number(10) NOT NULL,  
  DOI Date, DOR Date,  
  Constraint Date_Check Check (DOR>DOI);
```



Limitations of Check Constraint

ID	Name	Degree	Marks	DOB
1	Tim	BS	60	12-DEC-1980
2	Jim	MBA	70	13-JAN-1989
3	Cedric	MS	78	10-JUN-1980

Check Degree in ('BS','MS','MBA')

Limitations of Check Constraint

ID	Name	Degree	Marks	DOB
1	Tim	BS	60	12-DEC-1980
2	Jim	MBA	70	13-JAN-1989
3	Cedric	MS	78	10-JUN-1980
4	Rahat	BA	76	15-MAY-1980

Check Degree in ('BS','MS','MBA')
It will not be allowed...

Limitations of Check Constraint

ID	Name	Degree	Marks	DOB
1	Tim	BS	60	12-DEC-1980
2	Jim	MBA	70	13-JAN-1989
3	Cedric	MS	78	10-JUN-1980
4	Rahat	BA	76	15-MAY-1980



It is not desirable.
Its solution is
Foreign key
constraint

Check Degree in ('BE', 'ME', 'MCA')
To allow it we have to add it in check list
by changing schema

Limitations of Check Constraint

- ▶ It will not be suitable if check values need to be changed in future.
- ▶ Degree can be added or removed in future, for check constraint is not desirable.
- ▶ Apply check constraint only if its value is not going to change in future.
- ▶ For example,
 - ▶ Marks number(3) check (marks>=0)
 - ▶ Gender
Gender char(10) check (gender in ('male','female','transgender'));



Summary

- ▶ Importance of check constraint
- ▶ Its implementation and limitations

Next Session:

- ▶ Concept of foreign key and default constraint
- ▶ Recommended Readings:
 - ▶ More on Creation of tables with constraints in reading section of Week-2



Limitations of Check Constraint

- ▶ It will not be suitable if check values need to be changed in future.
- ▶ Degree can be added or removed in future, for check constraint is not desirable.
- ▶ Apply check constraint only if its value is not going to change in future.
- ▶ For example,
 - ▶ Marks number(3) check (marks>=0)
 - ▶ Gender
Gender char(10) check (gender in ('male','female','transgender'));



EmpID	EmpName	DeptID	Email
I01	Alice	D01	alice@email.com
I02	David	D02	david12@email.com
I03	Alice	D01	alice.dup@email.com
I04	David	D03	david@email.com



EmpID	EmpName	DeptID	Email
I01	Alice	D01	alice@email.com
I02	David	D02	david12@email.com
I03	Alice	D01	alice.dup@email.com
I04	David	D03	david@email.com

Department Table

D01	HR	New York
D02	Finance	Chicago
D03	IT	San Francisco



Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30

FK

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Only those values are allowed which exist in dno of Department table

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30
6	Rupinder	Prof	40

No corresponding department exists in University. Not allowed.

FK

P
K

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Deptno of Employee table refers to other table's PK for its validation, so it is known as FK



Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30
6	Rupinder	Prof	40

No corresponding department exists in University. Not allowed.

FK

P
K

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Deptno of Employee table refers to other table's PK for its validation, so it is known as FK



Foreign key Constraint

- ▶ Used to enforce referential integrity of data and establishes a relationship among tables.
- ▶ A foreign key may be a single column or the combinations of columns, which derive their values, based on the primary key of some other table.
- ▶ A table in which foreign key is present is known as child table and to which it refers is known as master or parent table.
- ▶ Its Syntax is:

Columnname datatype (size) references tablename [(columnname)]

[ON DELETE RESTRICT/ON DELETE CASCADE/ON DELETE SET NULL]



Syntax: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30

FK

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

1

Create table department
(deptno number(2) primary key,
dname char(20));

2

Create table employee
(eno number(2) primary key,
ename char(20), job char(10),
deptno number(2) references department(deptno));

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30

FK

P
K

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

What will happen if we try to delete deptno 30 record from department table with
Delete department where deptno=30;

It should not be allowed, Oracle achieves this by enforcing **ON DELETE RESTRICT** constraint on FK.

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30

FK

P
K

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

What will be the solution if we have to delete department number 30 from department table

Solution: Apply ON DELETE CASCADE constraint during creation of table so that department and corresponding employees records removed together.

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30

FK

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

Result of Delete department
where deptno=30 ;
will be

With ON DELETE CASCADE constraint

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20

Department table

Deptno	Dname
10	Computer
20	Chemical

FK

P
K

Record of department number
30 has been removed from
both tables.

With ON DELETE CASCADE constraint
corresponding employees have been fired

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	30

FK

P
K

Department table

Deptno	Dname
10	Computer
20	Chemical
30	Mechanical

What will be the solution if we have to delete department number 30 from department table, without firing corresponding employees?

Solution: Apply ON DELETE SET NULL
Corresponding employees records will not be removed and their dno will be set to NULL

Concept of Foreign Key: Case Study EMP-DEPT

Employee table

Eno	Ename	Job	Deptno
1	John	Prof	10
2	Smith	Asst Prof	20
3	Tom	Asst Prof	10
4	Davis	Prof	20
5	Emma	Asst Prof	

FK

Department table

Deptno	Dname
10	Computer
20	Chemical

Record of department number 30 has been removed from department table and corresponding employee record set to NULL

With ON DELETE SET NULL Constraint corresponding employees records has not been removed and set to NULL

1. Create a table **EMPLOYEE** without constraints:

- Columns: **Empno**, **Ename**, **Job**, **Sal**, **Deptno**.

2. Insert records into **EMPLOYEE**:

- Insert sample records (including duplicate **Empno**, NULL values, and invalid **Job** and **Deptno** entries).

3. Identify Problems:

- Identify potential problems in the data due to the lack of constraints (e.g., duplicate records, missing values, invalid entries).

Applying Constraints



Creating Master and Child tables with ON DELETE RESTRICT (By default)

- ▶ Parent Table
- ▶ SQL>CREATE TABLE department
(DEPTNO NUMBER(2) PRIMARY KEY,
DNAME CHAR(10));
- ▶ Child table
- ▶ SQL>CREATE TABLE employee(
empno NUMBER(4) PRIMARY KEY,
ename CHAR(20), job CHAR(10),
deptno NUMBER(2) REFERENCES department(deptno)
ON DELETE RESTRICT);



Creating Master and Child tables with ON DELETE CASCADE

- ▶ Parent Table
- ▶ SQL>CREATE TABLE department
(DEPTNO NUMBER(2) PRIMARY KEY,
DNAME CHAR(10));
- ▶ Child table
- ▶ SQL>CREATE TABLE employee(
empno NUMBER(4) PRIMARY KEY,
ename CHAR(20), job CHAR(10),
deptno NUMBER(2) REFERENCES department(deptno)
ON DELETE CASCADE);



Creating Master and Child tables with ON DELETE SET NULL

- ▶ Parent Table
- ▶ SQL>CREATE TABLE department
(DNO NUMBER(2) PRIMARY KEY,
DNAME CHAR(10));
- ▶ Child table
- ▶ SQL>CREATE TABLE employee(
empno NUMBER(4) PRIMARY KEY,
ename CHAR(20), job CHAR(10),
dno NUMBER(2) REFERENCES department(deptno)
ON DELETE SET NULL);



Concept of Foreign Key:

Case Study STUDENT-Degree

Rno	Name	Major	Marks	DOB
1	John	BS	60	12-DEC-1980
2	Smith	MBA	70	13-JAN-1989
3	Tom	MS	78	10-JUN-1980

How Foreign
key constraint
is implemented
in this case?



Concept of Foreign Key:

Case Study STUDENT-Degree

Student table

FK

Rno	Name	Degree	Marks
1	John	BE	80
2	Smith	MBA	89
3	Tom	MS	78
4	Davis	BA	78
5	Emma	BA	78

Degree table

Degree	Description
BS	Bachelor of Science
MS	Master of Science
MBA	Master of Business Administration

How we can add
BA Degree ?



Concept of Foreign Key:

Case Study STUDENT-Degree

Student table

FK

Rno	Name	Degree	Marks
1	Ram	BE	80
2	Rajesh	MBA	89
3	Suresh	MS	78
4	Ramesh	BA	78
5	Surinder	BA	78

P
K

Degree table

Degree	Description
BS	Bachelor of Science
MS	Master of Science
MBA	Master of Business Administration
BA	Bachelor of Arts

Add BA record in master Degree table and now students can be assigned to this degree

Concept of Foreign Key:

Case Study STUDENT-Degree

Student table

FK

Rno	Name	Degree	Marks
1	Ram	BE	80
2	Rajesh	ME	89
3	Suresh	BE	78
4	Ramesh	MCA	78
5	Surinder	MCA	78
6	Rupinder	MBA	56

P
K

Degree table

Degree	Description
BE	Bachelor of Engineering
ME	Master of Engineering
MCA	Master of Computer Applications
MBA	Master of Business Administration

Roll Number 6 has been assigned to MBA Degree and it satisfies referential integrity

Sno	Pno	Qty
S1	P1	100
S1	P2	100
S2	P3	100
S3	P1	100
S2	P2	1000





Sno	Pno	Qty
S1	P1	500
S1	P2	300
S2	P3	150
S3	P1	800
S2	P2	1000

Sno	Sname	City
S1	ABC Supplies	New York
S2	Global Hardware Co.	San Francisco
S3	Midwest Steelworks	Chicago
S4	Atlantic Materials	Miami



Sno	Pno	Qty
S1	P1	500
S1	P2	300
S2	P3	150
S3	P1	800
S2	P2	1000

Sno	Sname	City
S1	ABC Supplies	New York
S2	Global Hardware Co.	San Francisco
S3	Midwest Steelworks	Chicago
S4	Atlantic Materials	Miami

Pno	Pname	Color
P1	Bolt	Silver
P2	Washer	Black
P3	Nut	Bronze
P4	Screw	Zinc



USER_CONSTRAINTS and USER_CONS_COLUMNS tables to list constraints details

- ▶ To see the name of the constraints on a table you can query USER_CONSTRAINTS table (maintains internally by Oracle).

To query it:

- ▶ `SELECT * FROM USER_CONSTRAINTS WHERE
TABLE_NAME='NAME_OF_TABLE';`

For example:

- ▶ `SELECT * FROM USER_CONSTRAINTS WHERE
TABLE_NAME='STUDENT';`
- ▶ `SELECT * FROM USER_CONS_COLUMNS WHERE
TABLE_NAME='STUDENT';`



Summary

- ▶ Default Constraint
- ▶ USER_CONSTRAINTS and USER_CONS_COLUMNS tables to view information constraints applied on a table
- ▶ Recommended Readings:
 - ▶ More on Creation of tables with constraints in reading section of Week-2





Thanks